

Q1: Determine the critical forces for a rectangular plate with simply-supported edges, uniformly compressed in two directions, using the equilibrium method. Assume $q_x = \lambda$, $q_y = \alpha\lambda$, where $\alpha > 0$ is a fixed known parameter. Also, assume that the plate will buckle with one half-wave in the y -direction ($n = 1$).

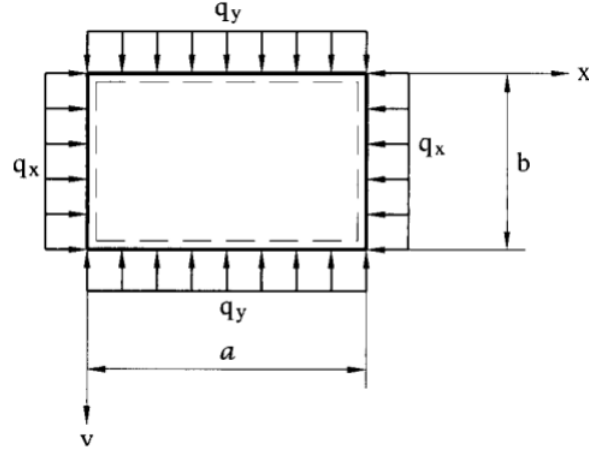


Figure 1: Q1's simply supported rectangular plate

Solution

For a thin plate subjected to in-plane compressive stress resultants N_x , N_y , and N_{xy} , the equilibrium equation governing the transverse deflection $w(x, y)$ is

$$\nabla^4 w = \frac{1}{D} (p + N_x w_{xx} + N_y w_{yy} + N_{xy} w_{xy}) \quad (1)$$

where

$$D = \frac{Eh^3}{12(1 - \nu^2)} \quad (2)$$

With $N_x = -q_x = -\lambda$, $N_y = -q_y = -\alpha\lambda$, and $N_{xy} = 0$, equation (1) becomes

$$D \nabla^4 w = -(\lambda w_{xx} + \alpha\lambda w_{yy}) \quad (3)$$

For a plate with all four edges simply supported, the boundary conditions are

$$w = 0, \quad \frac{\partial^2 w}{\partial x^2} = 0 \quad \text{at } x = 0 \text{ and } x = a \quad (4)$$

$$w = 0, \quad \frac{\partial^2 w}{\partial y^2} = 0 \quad \text{at } y = 0 \text{ and } y = b \quad (5)$$

According to boundary conditions we use Navier's approach for solve the problem. The deflection equation is

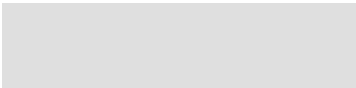
$$w(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (6)$$

Since the problem states that the plate buckles with $n = 1$ half-wave in the y -direction, we retain a single term as

$$w(x, y) = A_m \sin \frac{m\pi x}{a} \sin \frac{\pi y}{b} \quad (7)$$

Which satisfies all four boundary conditions.

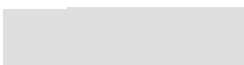
– $w = 0$ at $x = 0, a$ and $y = 0, b$

–  at $x = 0, a$

–  at $y = 0, b$

Now, We compute all required derivatives of (7).

$$\text{} \quad (8)$$

$$\text{} \quad (9)$$

$$\text{} \quad (10)$$

$$\text{} \quad (11)$$

$$\text{} \quad (12)$$

Therefore the biharmonic operator gives

$$\text{} \quad (13)$$

Substituting (8), (9), and (13) into (3):

$$\text{} \quad (14)$$

One answer is $w = 0$ which relate to the initial stability. The non-trivial solution is obtained by setting the determinant of the coefficient matrix to zero.

$$\text{} \quad (15)$$

Dividing throughout by $w \neq 0$ (non-trivial buckling mode) and factoring out π^2

$$\text{} \quad (16)$$

Solving (16) for λ

$$\lambda_m = \frac{D\pi^2}{b^2} \cdot \frac{\left(\frac{m^2}{r^2} + 1\right)^2}{\frac{m^2}{r^2} + \alpha} \quad r = \frac{a}{b}, \quad m = 1, 2, 3, \dots \quad (17)$$

Introducing the plate aspect ratio $r = a/b$ and multiplying numerator and denominator by b^4

$$\lambda_m = \frac{D\pi^2}{b^2} \cdot \frac{\left(\frac{m^2}{r^2} + 1\right)^2}{\frac{m^2}{r^2} + \alpha} \quad r = \frac{a}{b}, \quad m = 1, 2, 3, \dots \quad (18)$$

The critical load is the *minimum* value of λ_m over all admissible integers $m \geq 1$

$$\lambda_{cr} = \min_{m=1,2,3,\dots} \lambda_m \quad (19)$$

To identify the minimizing m , we treat it temporarily as a continuous variable. Let $u = m^2/r^2 \geq 0$, so that

$$\lambda_m = \frac{D\pi^2}{b^2} \cdot \frac{(u + 1)^2}{u + \alpha} \quad (20)$$

Setting $\frac{d\lambda}{du} = 0$

$$\frac{d}{du} \left(\frac{(u + 1)^2}{u + \alpha} \right) = 0 \quad (21)$$

The numerator must vanish (and $u + 1 \neq 0$)

$$\frac{d}{du} \left(\frac{(u + 1)^2}{u + \alpha} \right) = \frac{2(u + 1)(u + \alpha) - (u + 1)^2}{(u + \alpha)^2} = 0$$

$$m^* = r\sqrt{1 - 2\alpha} \quad (22)$$

Two cases arise:

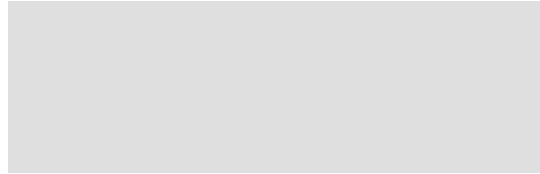
- Case $\alpha < \frac{1}{2}$: A real positive minimizer $m^* > 0$ exists from (22). Round m^* to the nearest positive integer, evaluate λ_m for $m = \lfloor m^* \rfloor$ and $m = \lceil m^* \rceil$, and select the smaller value.
- Case $\alpha \geq \frac{1}{2}$: No real positive minimizer exists; λ_m is monotonically decreasing in m for all $m \geq 1$, so $m = 1$ always gives the smallest value.

$$\lambda_{cr} = \min_{m=1,2,3,\dots} \left\{ \frac{D\pi^2}{b^2} \cdot \frac{\left(\frac{m^2}{r^2} + 1\right)^2}{\frac{m^2}{r^2} + \alpha} \right\}, \quad r = \frac{a}{b} \quad (23)$$

Let's check three cases:

Case 1: Uniaxial Compression ($\alpha = 0$, only $q_x = \lambda$)

The denominator of (18) reduces to m^2/r^2 and the formula becomes


(24)

The minimum occurs at $m^* = r$,

$$\lambda_{cr} = \frac{4D\pi^2}{b^2} = \frac{\pi^2 E h^3}{3b^2(1 - \nu^2)} \quad (25)$$

Case 2: Equal Biaxial Compression ($\alpha = 1$, $q_x = q_y = \lambda$)

For $m = 1$ and $n = 1$,

$$\lambda_{cr} = D\pi^2 \left(\frac{1}{a^2} + \frac{1}{b^2} \right) = \frac{\pi^2 E h^3}{12(1 - \nu^2)} \left(\frac{1}{a^2} + \frac{1}{b^2} \right) \quad (26)$$

Case 3: Square Plate with Equal Biaxial Compression ($a = b$, $\alpha = 1$) For $a = b$ with $m = 1$ and $n = 1$,

$$\lambda_{cr} = \frac{2D\pi^2}{a^2} = \frac{\pi^2 E h^3}{6a^2(1 - \nu^2)} \quad (27)$$

Q2: Let a rectangular plate of sides $2a$ and $2b$ ($a > b$) with fixed edges be subjected to compressive linearly varying loads

$$q_x = q_0 \left(1 - \frac{\eta y}{b}\right) \quad (\eta < 1). \quad (28)$$

The load is applied to short plate edges. The origin of the Cartesian coordinate system is attached at the midpoint of the plate. Determine the critical value of the applied compressive load by the Ritz method. Assume that

$$2a = 3 \text{ m}, \quad 2b = 1.5 \text{ m}, \quad t = 0.15 \text{ m}, \quad \eta = 0.5, \quad E = 200 \text{ GPa}, \quad \text{and} \quad \nu = 0.28.$$

Consider the approximate expression for deflection as follows:

$$w(x, y) = C(x^2 - a^2)^2(y^2 - b^2)^2. \quad (29)$$

Solution

For a fixed plate, the boundary conditions are

$$w = 0, \quad \frac{\partial w}{\partial x} = 0 \quad \text{at } x = \pm a \quad (30)$$

$$w = 0, \quad \frac{\partial w}{\partial y} = 0 \quad \text{at } y = \pm b \quad (31)$$

Calculate partial differential of w :

$$\frac{\partial w}{\partial x} = 4C(x^2 - a^2)(y^2 - b^2)^2 x \quad (32)$$

$$\frac{\partial w}{\partial y} = 4C(x^2 - a^2)^2(y^2 - b^2)y \quad (33)$$

$$\frac{\partial^2 w}{\partial x^2} = 4C(y^2 - b^2)^2(3x^2 - a^2) \quad (34)$$

$$\frac{\partial^2 w}{\partial y^2} = 4C(x^2 - a^2)^2(3y^2 - b^2) \quad (35)$$

$$\frac{\partial^2 w}{\partial x \partial y} = 8C(x^2 - a^2)(y^2 - b^2)x y \quad (36)$$

Which satisfies all four boundary conditions.

– $w = 0$ at $x = \pm a$, and $y = \pm b$

– $\frac{\partial w}{\partial x} = 0$ at $x = \pm a$

$$- \quad \text{[redacted]} \quad \text{at } y = \pm b$$

Variation of total potential energy equation is

$$\begin{aligned} \Delta \Pi &= \Delta U + \Delta V \\ &= \frac{1}{2} \int_{-b}^b \int_{-a}^a \left[N_x \left(\frac{\partial w}{\partial x} \right)^2 + N_y \left(\frac{\partial w}{\partial y} \right)^2 + N_{xy} \left(\frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \right) \right] dx dy \\ &\quad + \frac{D}{2} \int_{-b}^b \int_{-a}^a \left[\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right)^2 - 2(1 - \nu) \left(\frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} - \left(\frac{\partial^2 w}{\partial x \partial y} \right)^2 \right) \right] dx dy \end{aligned} \quad (37)$$

Consider $N_x = -q_x$ ((28)), $N_y = 0$, $N_{xy} = 0$ and substituting (32)–(36) into (37).

With $N_y = 0$, $N_{xy} = 0$,

$$\text{[redacted]} \quad (38)$$

x -integral:

$$\text{[redacted]} \quad (39)$$

y -integral (the antisymmetric part $\frac{\eta y}{b}(y^2 - b^2)^4$ vanishes by symmetry):

$$\text{[redacted]}$$

Define the following auxiliary integrals:

$$\text{[redacted]} \quad (40)$$

$$\text{[redacted]} \quad (41)$$

$$\text{[redacted]} \quad (42)$$

Similarly for y -integrals:

$$\text{[redacted]} \quad (43)$$

$$\text{[Redacted]} \quad (44)$$

$$\text{[Redacted]} \quad (45)$$

$(w_{xx} + w_{yy})^2$ term:

$$\text{[Redacted]} \quad (46)$$

Integrating term by term:

$$\text{[Redacted]} \quad (47)$$

$w_{xx}w_{yy} - w_{xy}^2$ term:

$$\text{[Redacted]} \quad (48)$$

$$\text{[Redacted]} \quad (49)$$

$$\text{[Redacted]} \quad (50)$$

$$\text{[Redacted]} \quad (51)$$

Since both integrals are equal

$$\text{[Redacted]} \quad (52)$$

Substituting obtained values into (37),

$$\text{[Redacted]} \quad (53)$$

Applying the Ritz condition $\frac{\partial(\Delta\Pi)}{\partial C} = 0$ (non-trivial solution, $C \neq 0$),

$$\text{[Redacted]} \quad (54)$$

